Executive Summary: Cloud, Persistence & Sync

This audit report evaluates the MEP Calculator codebase against the FENMO (Globel) Application QA/UAT Checklist for Cloud Storage, Real-Time Collaboration & Data Integrity (IDs 6, 7, 16, and 30). It validates persistence across sessions, multi-user WebSocket synchronization, and crash recovery protections.

ID	Checklist Item	Severity	Audit Result	Action Required
ID 6	Save to Cloud (Permanent Cloud Persistence)	High	PASS (100%)	None - Supabase cloud storage & autosave active
ID 7	Auto Sync (Multi-Client Real-Time Collaboration)	High	PASS (100%)	None - Supabase Realtime WebSocket operational
ID 16	Server Storage / Permanent Retention	High	PASS (100%)	None - Dual-tier cloud DB & local disk backups active
ID 30	Data Integrity (Loss Prevention & Sanitization)	High	PASS (100%)	None - Undo/Redo & input sanitizers operational

1. Detailed Architecture Verification & Findings

Item ID	Architecture Component	Technical Verification & Evidence
ID 6	Cloud Save Pipeline File: frontend/src/lib/workspace-store.ts Endpoint: POST /api/projects/save-workspace	EVIDENCE: `workspaceStore.forceSaveToCloud()` and debounced autosave (1000ms timer) serialize current spaces, zones, equipment, and form caches to Supabase PostgreSQL cloud table `projects`. On page reload or fresh re-login, `syncFromCloud()` reliably pulls and hydrates the exact workspace state.
ID 7	Real-Time Collaboration Engine File: frontend/src/lib/collaboration-store.ts Protocol: Supabase Realtime Channels	EVIDENCE: Subscribes to `project:${projectName}` channel with presence tracking and broadcast channels. Form updates in `AppShell.tsx` trigger `broadcastData(pathname, state)`, which dispatches DOM updates to other engineers currently viewing the same project room.
ID 16	Permanent Retention Strategy File: backend/authServer.js Storage: Cloud PostgreSQL + Local JSON	EVIDENCE: Dual-tier redundancy architecture. Primary storage resides permanently in cloud database without expiration or TTL eviction. Server-side auto-backups are simultaneously mirrored to `backend/data/` JSON stores for instant disaster recovery.
ID 30	Data Integrity & Crash Safety File: frontend/src/lib/workspace-store.ts Modules: sanitizeWorkspaceState & useAutoBackup	EVIDENCE: Pre-save validation sanitizes empty names, resolves orphaned zones to Default, and validates calculation parameters. Includes 50-step in-memory Undo/Redo stack (`pushHistory()`) and periodic snapshot timers (`useAutoBackup`).

2. Multi-Tier Persistence & Sync Pipeline Diagram

The following diagram outlines the end-to-end cloud persistence, real-time collaboration, and local caching architecture:

1. Local Client Layer (Instant Interaction & Crash Safety)

• React State Store (Zustand + workspaceStore) maintains active spaces, zones, and equipment.
• LocalStorage cache (`mep_cached_projects`, `mep_auto_state_*`) provides instant zero-delay boot.
• 50-step Undo/Redo historical stack captures structural changes prior to save triggers.

2. Real-Time Collaboration Layer (WebSocket Multi-Client Sync)

• Supabase Realtime WebSocket establishes bidirectional channel `project:<id>`.
• Dynamic presence broadcast detects peer engineers and displays active user avatar indicators.
• Form input changes broadcast live DOM state to other collaborators in real-time.

3. Cloud & Server Permanent Retention Layer

• Debounced (1000ms) and explicit `forceSaveToCloud()` dispatches `POST /api/projects/save-workspace`.
• Database persistence in Supabase PostgreSQL `projects.workspace_data` (JSONB column).
• Server-side mirror write to local filesystem `backend/data/` for resilient offline continuity.

3. Audit Conclusion & Compliance Status

All 4 checklist requirements (IDs 6, 7, 16, and 30) under Cloud, Persistence & Synchronization are fully verified, robust, and 100% compliant with enterprise cloud storage and multi-user standards. No code modifications are required for this module.